

Frozen Lake from First Principles using Q-Learning

Technical Report — DSCD 614 Reinforcement Learning, Assignment 1

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1. Introduction

This report explains my solution to the 8×8 Frozen Lake problem using

In Reinforcement Learning an agent learns by trying actions in an environment.

2. Environment Design

I wrote the environment as a class called FrozenLakeEnv with the methods the

- States: I used single numbers from 0 to 63. To get the grid position I do

The class keeps track of the current state, checks the boundaries, finds holes

3. Q-Learning Algorithm

Q-Learning keeps a table $Q(s,a)$ that estimates how good it is to take action

$$Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \cdot \max_{a'} Q(s',a') - Q(s,a)]$$

Here α is the learning rate (how fast it learns), γ is the discount factor

For choosing actions I used ϵ -greedy: with probability ϵ the agent picks a

4. Training Methodology

The training loop is in train.py. It plays many episodes, updates the Q-table

Parameter	Value	Parameter	Value
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I tried a few different values for the learning rate, discount and exploration.

5. Experimental Results

When I tested the trained agent greedily for 100 episodes on the normal

Bonus B (graphs): the reward, success-rate and epsilon graphs are made

6. Learned Policy

The policy I got from the Q-table (Part D) avoids the holes in the middle and

↓	↓	↓	↓	↓	↓	↓	↓
→	→	→	→	→	↓	↓	↓
↑	↑	↑	H	→	→	→	↓
↑	↑	↑	H	↑	→	→	↓
↑	↑	↑	H	→	→	→	↓
↑	H	H	→	↑	↑	H	↓
←	H	←	←	H	↑	H	↓
←	←	←	H	←	←	←	G

Legend: ← Left, ↓ Down, → Right, ↑ Up, H Hole, G Goal. Some cells that are not on

7. Challenges Encountered

- Reward only at the goal: because the reward only comes at the end, the

8. Conclusion

My from-scratch Q-Learning agent solves the normal 8×8 Frozen Lake perfectly

(100%)